Chengwei Liang

Personal Information

Name: Chengwei Liang
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Education

Bachelor of Science, Guangzhou University, China *Sept. 2023 – July. 2027 (expected)*
Major: Astronomy; GPA: 3.24/4.00
Supervisor: Rui Luo (GZHU)

Research Interests

- FRB & Pulsar searching and polarization measurement.
- Machine learning and deep learning in the development of FRB classification and searching pipeline.

Publications

First-Author Papers

1. **Chengwei Liang**, Shiqian Zhao, Rui Luo, Yuanpei Yang et al., Targeted Search for Fast Radio Bursts from the Young Pulsar Wind Nebula Candidate VT 1137−0337 (Manuscript prepared for submission)

Others

2. Junshuo Zhang(NAOC), Yongkun Zhang(NAOC), Yi Feng(Zhejiang Lab), **Chengwei Liang(GZHU)**, Shuo Cao(NAOC) et al., Detection of high activity from repeating FRB 20121102A at L-band with FAST, *ATel #17642*.
3. Junshuo Zhang(NAOC), Dejiang Zhou(NAOC), Yi Feng(Zhejiang Lab), **Chengwei Liang(GZHU)**, Chenyuan Xu (Zhejiang Lab) et al., FRB 20121102A is exhibiting hyperactivity with increased RM value at L-band detected by FAST, *ATel #17921*.
4. S. Q. Zhao, X. C. Wu, R. Luo, S. Q. Wang et al. (including **Chengwei Liang**), Magnetic dipolar geometry unveiled from a long-period radio pulsar, preprint.
5. Jiarui Niu, Xiaohui Liu, Nan Xu, Songyu Shen et al. (including **Chengwei Liang**), No Strong Evidence for Plasma Lensing in FRB 20240114A, *ApJL* accepted for publication.
6. Xiaohui Liu, Jiarui Niu, Tiancong Wang, Jun-Shuo Zhang et al. (including **Chengwei Liang**), Random Polarization Position Angle Behaviors across Bursts of Repeating Fast Radio Bursts, *ApJ* 1006, 157 (2026).

Observational Programs

FAST:

- **PI:** Depicting the dynamic environment of FRB 20180301A with FAST (PT2026_0102): **3.5 hours** *2026 - 2027*
- **Co-I:** Observing a Peculiar Radio Transient with FAST (PT2026_0196): **3 hours** *2026 - 2027*

- **Co-I:** Depicting the dynamic environment of FRB 20180301A with FAST (PT2025_0015): **10 hours** *2025 - 2026*
- **Co-I:** Searching for fast radio bursts from a persistent radio source with likely magnetar origin (PT2024_0126): **5 hours** *2024 - 2025*

Parkes:

- **PI:** Depicting the dynamic environment of FRB 20180301A (P1328): **60 hours** *2026OCT*
- **PI:** Follow-up of FRB 20180301A (ToO; PX153): **21.5 hours** *Apr. 2026*
- **PI:** Probing the dynamic magneto-ionic environment of a high-RM repeating fast radio burst (P1409): **24 hours** *2026APR*
- **Co-I:** Searching for single pulses from an ultra-long period radio pulsar (P1327): **81.5 hours** in total *2025APR - 2025OCT*
- **Co-I:** Studying wideband spectro-polarimetric behaviour of FRB 20180301A (P1328): **71 hours** in total *2024OCT - 2025OCT*
- **Co-I:** Searching for fast radio bursts from a radio source with likely magnetar origin (P1281): **62 hours** in total *2023APR - 2024OCT*

Talks and Posters

Poster, FRB2026, Guiyang, China *Jul. 2026*
Unraveling the Spectro-polarimetric Difference of Repeating and Non-repeating FRBs

Talk, FAST/Future Pulsar Symposium 15, Guiyang, China *Jul. 2026*
Unraveling the Spectro-polarimetric Difference of Repeating and Non-repeating FRBs

Talk, Australasian "Orange" Pulsar Workshop 2026, Australia *May 2026*
Unraveling the polarimetric difference of Repeating and Non-repeating FRBs

Poster, Symposium on Fast Radio Bursts and Their Astrophysics, Beijing, China *Jun. 2025*
A targeted search for fast radio bursts from a radio source with likely magnetar origin

Professional Skills

- **FRB Search and Data Reduction**: Experienced in FRB searching and data reduction using TransientX, PRESTO, and related pipelines.
- **Radio Telescope Observations**: Certified Parkes Telescope Observer (Oct. 2025); experienced in conducting observations with Parkes and FAST.
- **Machine Learning and Deep Learning**: Experienced in applying ML/DL techniques to large-sample FRB analysis; NVIDIA Deep Learning Fundamentals – Theory and Practice certification (May 2026).

Collaboration

- FAST FRB KSP Collaboration, China *Jun. 2025 - now*
- ASKAP CRAFT Collaboration, Australia *Aug. 2026 - now*

References

Prof. Rui Luo Guangzhou University